Choose Informative Questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One current question-choice record, with earlier tree drafts preserved.

State possibilities, compare questions, recount a restricted set and explain assumptions and ties.

Keep one habit: Current tasks have separate answer fields; earlier tree questions and saved responses remain labelled review.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Decision Trees

Accuracy as a Fraction

Features and Distractors

W1 · State the current target possibilities

> Keep this current game record separate from the earlier tree task and saved response.

Current cards and criterion

Card	Subject	Binding	Language
M1	mystery	paperback	English
M2	mystery	paperback	English
M3	mystery	paperback	English
M4	mystery	hardback	English
S1	science	paperback	English
S2	science	paperback	English
S3	science	paperback	English
S4	science	hardback	English

Current game policy: one target among the current cards; answers use the printed fields and are truthful for the exercise. Compare mystery?, paperback?, English? in that order. Minimise max(Yes count,No count); keep the first if worst-case counts tie. Every current card enters one answer branch. This is an identification game, not the retained fitted-tree review.

Current task and new answer field

- Using the supplied eight-card table, list all possible target IDs. State what a truthful Yes or No means for the remaining set, and why English? cannot narrow the original set.

My current question-choice reasoning and evidence

▼ **Week 14 review · Earlier tree task and saved response**

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree.
Which parts were selected using **training** labels?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compare all three questions

> Keep this current game record separate from the earlier tree task and saved response.

Current cards and criterion

Card	Subject	Binding	Language
M1	mystery	paperback	English
M2	mystery	paperback	English
M3	mystery	paperback	English
M4	mystery	hardback	English

Card	Subject	Binding	Language
S1	science	paperback	English
S2	science	paperback	English
S3	science	paperback	English
S4	science	hardback	English

Current game policy: one target among the current cards; answers use the printed fields and are truthful for the exercise. Compare mystery?, paperback?, English? in that order. Minimise max(Yes count,No count); keep the first if worst-case counts tie. Every current card enters one answer branch. This is an identification game, not the retained fitted-tree review.

Current task and new answer field

- For mystery?, paperback? and English?, count Yes and No, calculate the largest remaining group and select the best under the stated criterion. Explain why one useful question need not identify a unique target.

My current question-choice reasoning and evidence

▼ **Week 14 review · Earlier tree task and saved response**

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Recount after mystery gets Yes

> Keep this current game record separate from the earlier tree task and saved response.

Current cards and criterion

Card	Subject	Binding	Language
M1	mystery	paperback	English
M2	mystery	paperback	English
M3	mystery	paperback	English
M4	mystery	hardback	English
S1	science	paperback	English
S2	science	paperback	English
S3	science	paperback	English
S4	science	hardback	English

Current game policy: one target among the current cards; answers use the printed fields and are truthful for the exercise. Compare mystery?, paperback?, English? in that order. Minimise max(Yes count,No count); keep the first if worst-case counts tie. Every current card enters one answer branch. This is an identification game, not the retained fitted-tree review.

Current task and new answer field

- Use only M1–M4 from the supplied table. Recount all three questions, choose the best now and explain why repeating mystery? no longer helps.

My current question-choice reasoning and evidence

▼ **Week 14 review · Earlier tree task and saved response**

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit the claim and explain a tie

> Keep this current game record separate from the earlier tree task and saved response.

Current cards and criterion

Card	Subject	Binding	Language
M1	mystery	paperback	English
M2	mystery	paperback	English
M3	mystery	paperback	English
M4	mystery	hardback	English
S1	science	paperback	English
S2	science	paperback	English
S3	science	paperback	English
S4	science	hardback	English

Current game policy: one target among the current cards; answers use the printed fields and are truthful for the exercise. Compare mystery?, paperback?, English? in that order. Minimise max(Yes count,No count); keep the first if worst-case counts tie. Every current card enters one answer branch. This is an identification game, not the retained fitted-tree review.

Current task and new answer field

- Does the smallest worst-case group guarantee truthful answers or minimum expected cost? State one assumption and limitation. If two questions have the same largest group, which does this exercise keep, and what does that convention fail to prove?

My current question-choice reasoning and evidence

▼ **Week 14 review · Earlier tree task and saved response**

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Changed independent case

Keep one first attempt before hints or feedback.

Read and explain independently

Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Ask first about preferences that remove the largest number of unsuitable books.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Verify the candidate scores for **Fitting** a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Eight blue cards contain four circles and four triangles; target is shape.

Which question is more useful: blue or circle?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 15 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Extra practice · Earlier tree-fitting transfer

REFERENCE · SECTION 15

Extra practice · Earlier tree-fitting transfer

> This is a retained Week 14-style task, not a duplicate of the selected ten-card W4.

Earlier task and saved response

- New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

  Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

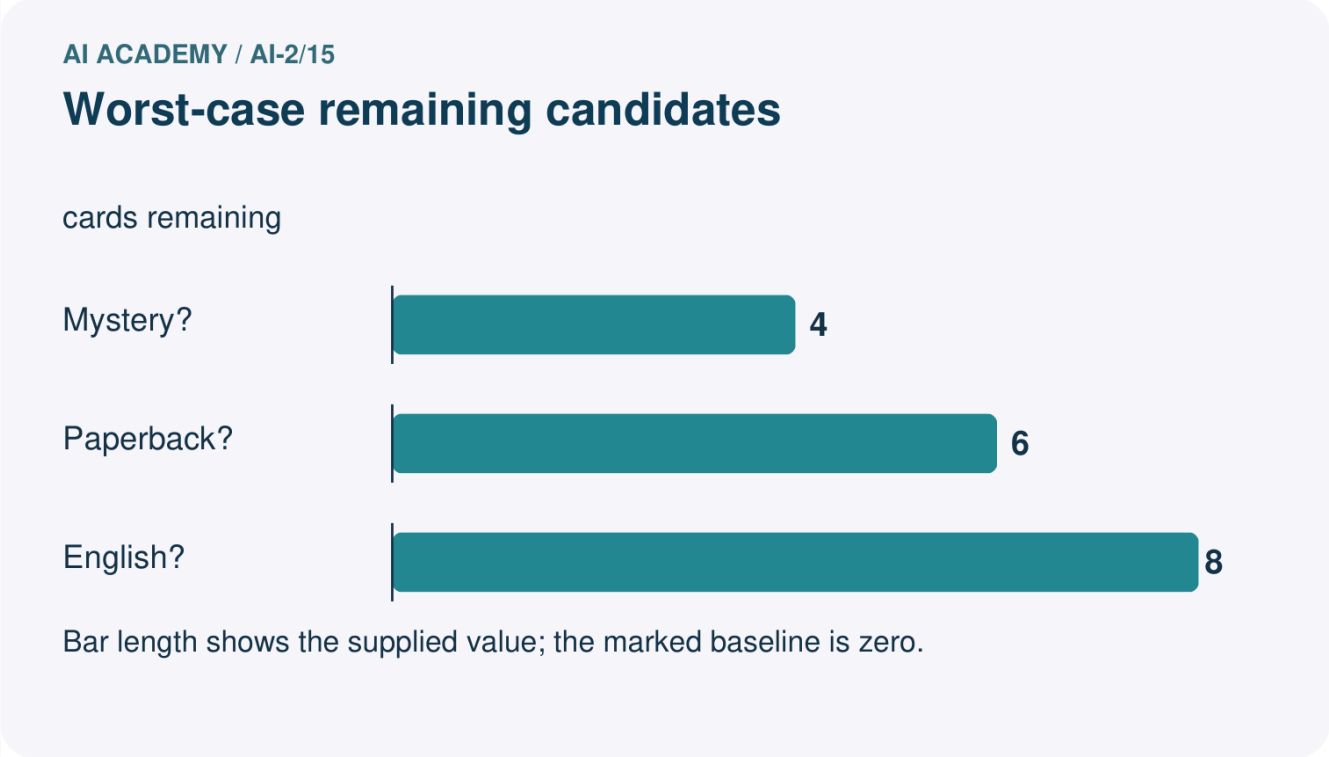
> Keep the explanation and the support you used.

Use this when helpful

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/15

Choose Informative Questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. List possibilities

There are eight possible target cards initially.



2. Mystery question

Yes group 4; No group 4; largest remaining group 4.



3. Paperback question

Yes 6; No 2; largest remaining group 6.



4. English question

Yes 8; No 0; largest remaining group 8.

CHANGE ONE THING

The current possibility set contains only the four mystery cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Choose the best original question and explain why its more balanced **split** matters here.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.
- Which labelled step matches this detail? Yes 8; No 0; largest remaining group 8.
- 1. List possibilities
- 2. Mystery question
- 3. Paperback question
- 4. English question
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: English question Yes 8; No 0; largest remaining group 8.

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Changed-case question

- Changed-case practice: The current possibility set contains only the four mystery cards. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Asking mystery now leaves all four; it no longer narrows the set.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Asking mystery now leaves all four; it no longer narrows the set. Question value depends on the remaining candidates, not just on its wording.

Repair and save your work

Review Decision Trees (ai-2/14). Goal: Fit a **root** question and **leaf prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

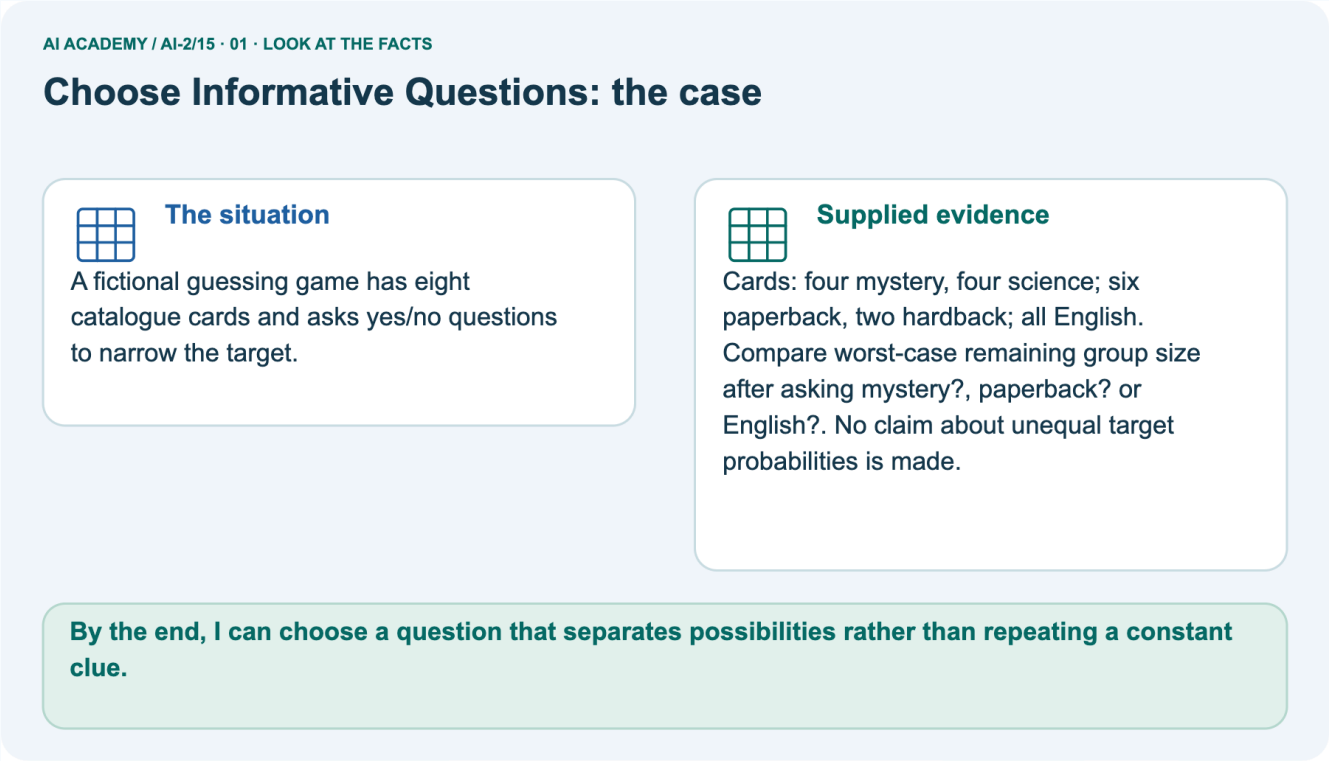
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the image that helps rather than repeating all six.

01. Choose Informative Questions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

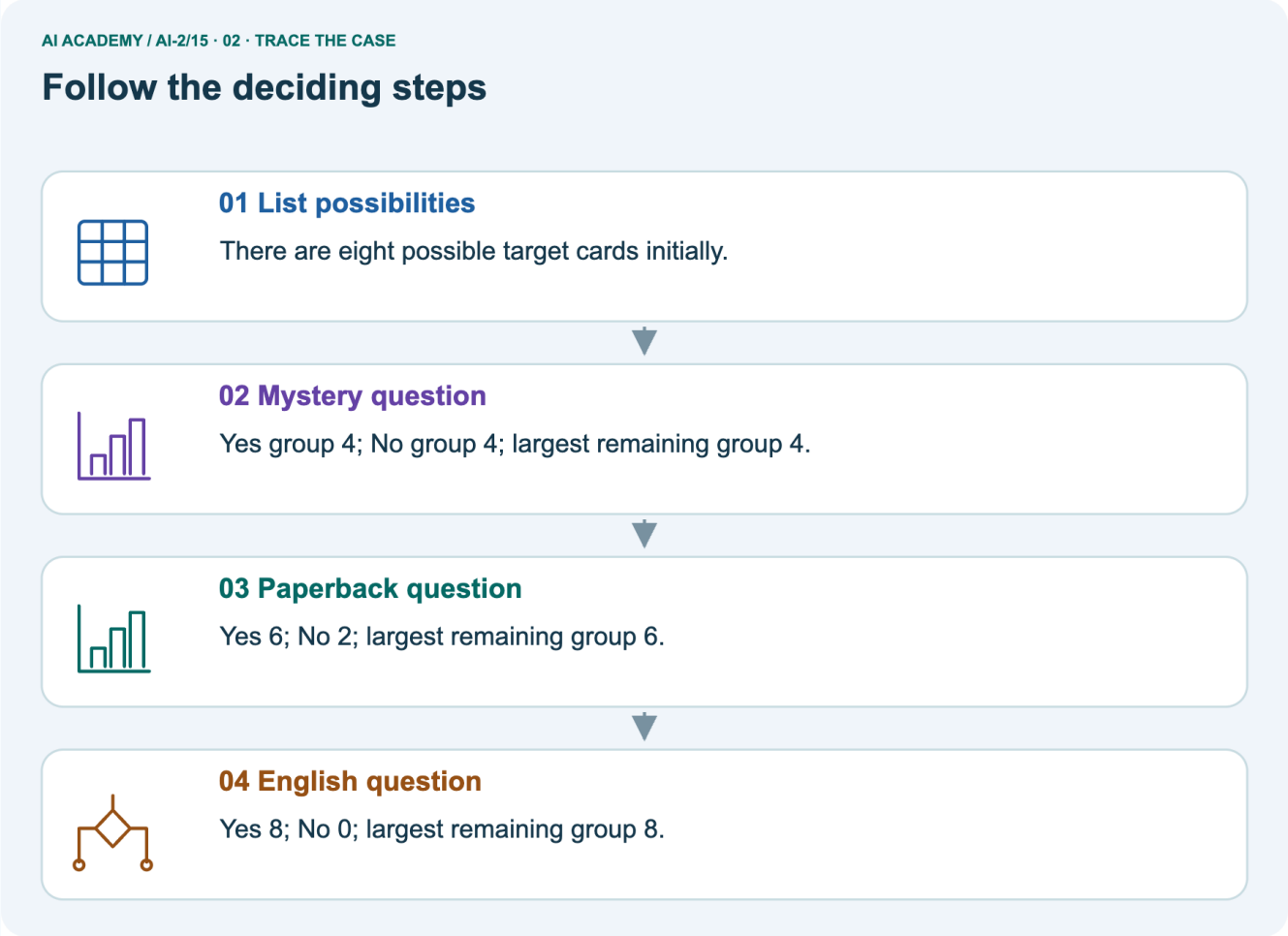
A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target. Supplied facts: Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



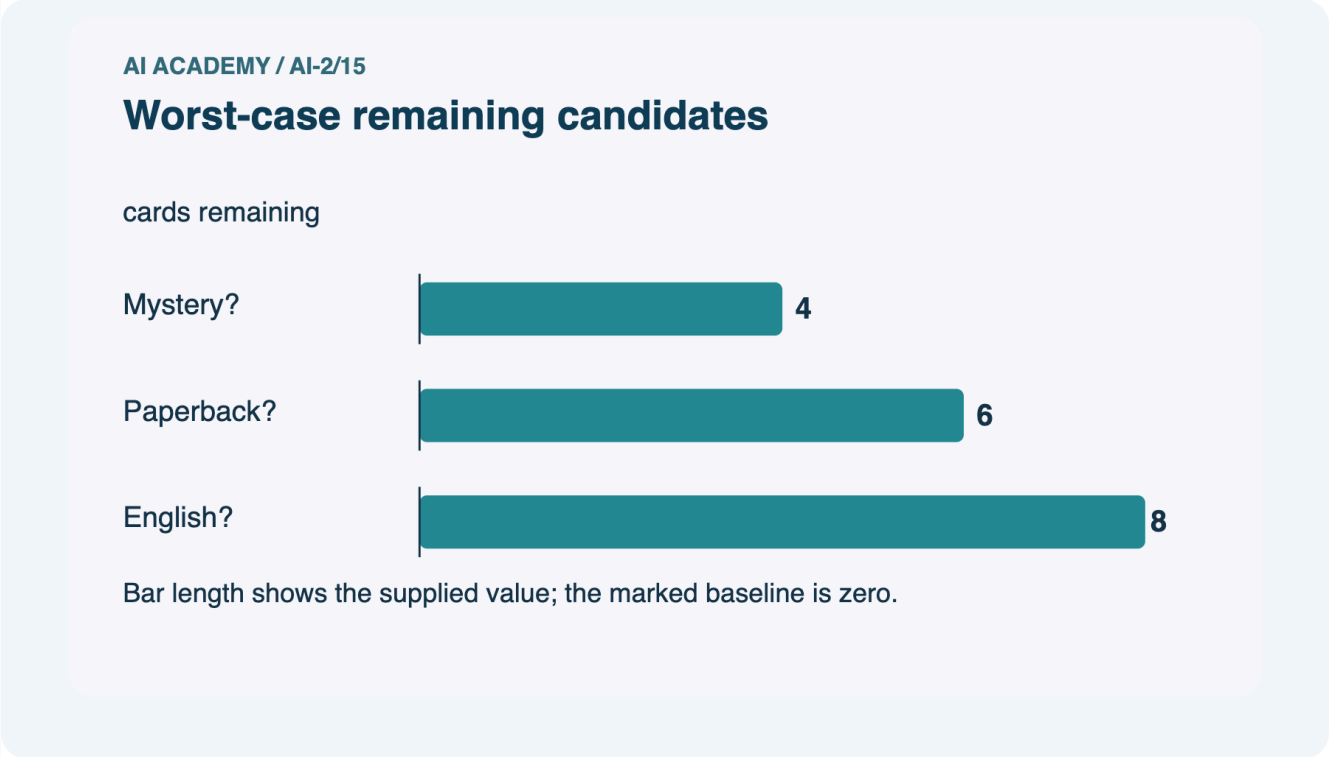
Follow the labelled picture with the matching explanation. Open the vector diagram

List possibilities: There are eight possible target cards initially. → Mystery question: Yes group 4; No group 4; largest remaining group 4. → Paperback question: Yes 6; No 2; largest remaining group 6. → English question: Yes 8; No 0; largest remaining group 8.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Worst-case remaining candidates



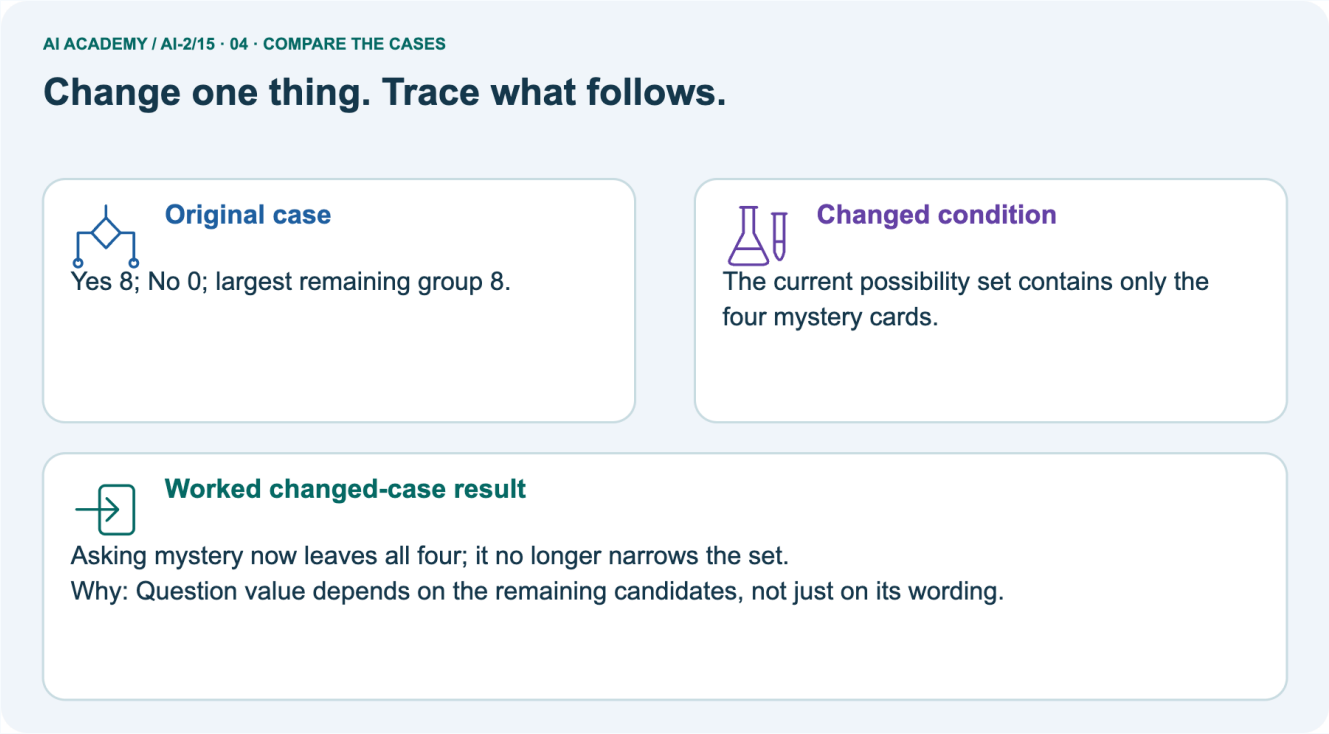
Follow the labelled picture with the matching explanation. Open the vector diagram

Worst-case remaining candidates. Units: cards remaining. Mystery?: 4. Paperback?: 6. English?: 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

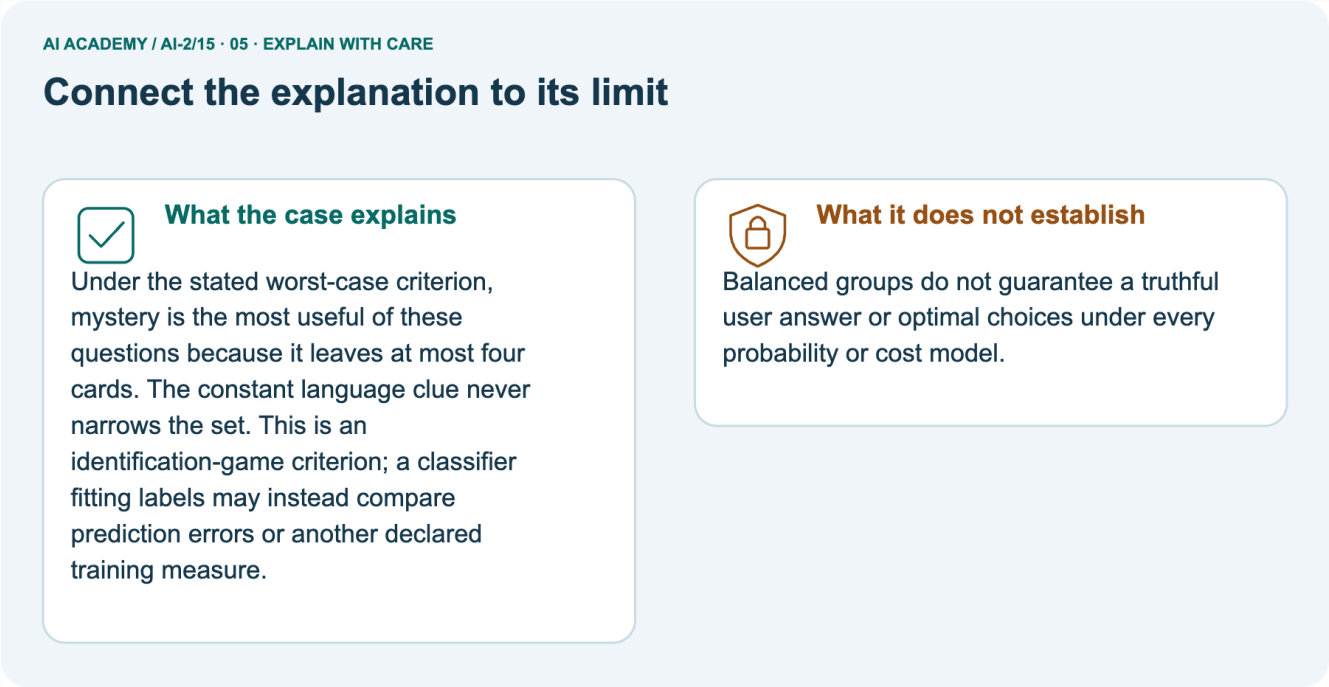
Original case: Yes 8; No 0; largest remaining group 8. Changed condition: The current possibility set contains only the four mystery cards. Expected result: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

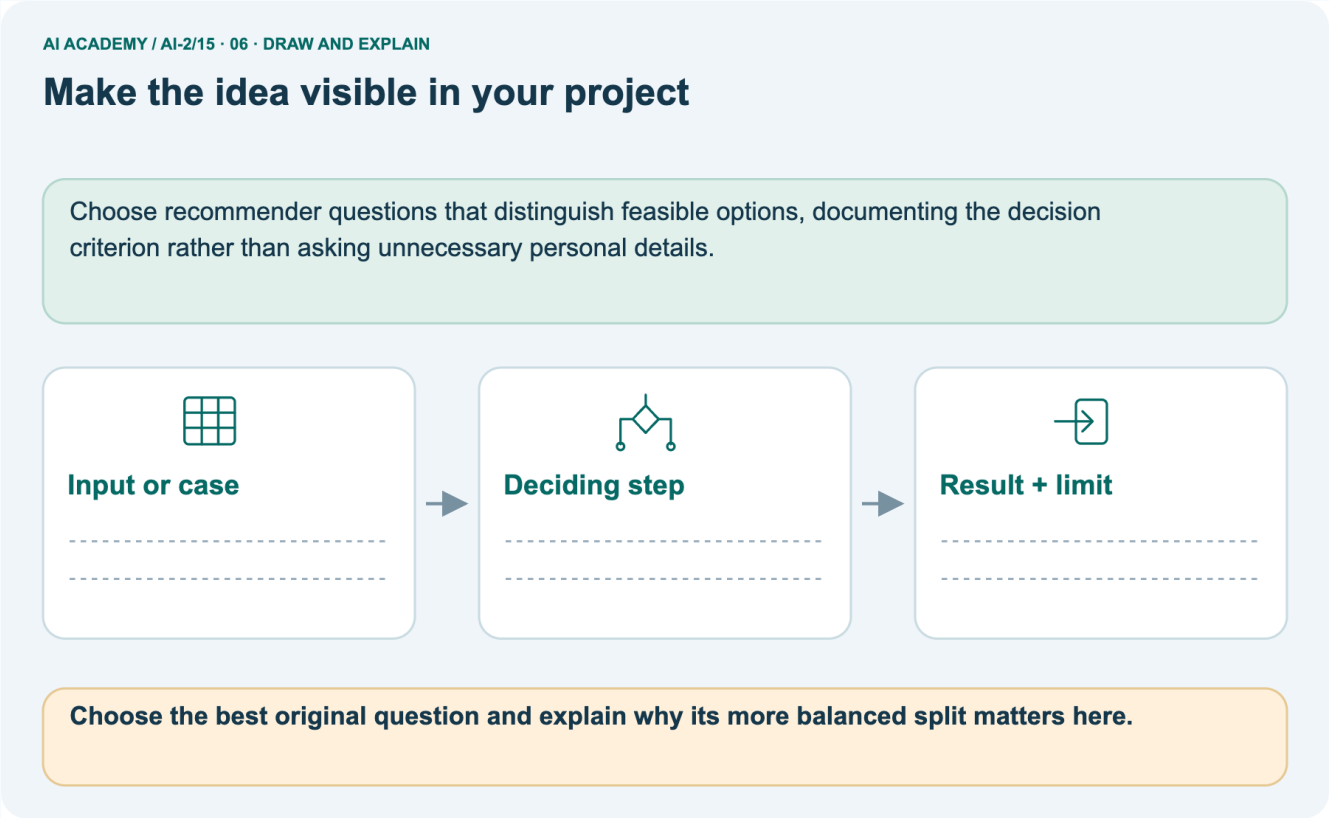
This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Limit: Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details. Draw your **input**, deciding step, result and limit.

Supported question: Choose the best original question and explain why its more balanced **split** matters here.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Choose Informative Questions: the case

02. Follow the deciding steps

03. Worst-case remaining candidates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.